

DISEASE/FINDING	MOST COMMON/IMPORTANT ASSOCIATIONS	PAGE
Kidney stones (radiolucent)	Uric acid	617
Kidney stones (radiopaque)	Calcium (most common), struvite (ammonium), cystine (faintly radiopaque)	617
Renal malignancy (in males)	Renal cell carcinoma: associated with tobacco smoking and VHL (clear cell subtype); may present with paraneoplastic syndromes (EPO, renin, PTHrP, ACTH)	623
1° amenorrhea	Turner syndrome (45,XO or 45,XO/46,XX mosaic)	655
Hypogonadotropic hypogonadism with anosmia	Kallmann syndrome (neuron migration failure)	656
Clear cell adenocarcinoma of the vagina	DES exposure in utero	662
Ovarian tumor (benign, bilateral)	Serous cystadenoma	664
Ovarian tumor (malignant)	Serous carcinoma	664
Benign tumor of myometrium	Leiomyoma (estrogen sensitive, not precancerous)	666
Gynecologic malignancy (most common)	Endometrial carcinoma (most common in resource-rich countries); cervical cancer (most common worldwide)	663–666
Breast mass	Fibrocystic change (in premenopausal females); carcinoma (in postmenopausal females)	667, 668
Breast tumor (benign, young woman)	Fibroadenoma	667
Breast cancer	Invasive ductal carcinoma	668
Bloody nipple discharge	Intraductal papilloma (no palpable mass); invasive ductal carcinoma (hard palpable mass)	
Testicular tumor	Seminoma (malignant, radiosensitive, ↑ placental alkaline phosphatase [PLAP])	670, 671
Bladder outlet obstruction in men	BPH	672
Pulmonary hypertension	Idiopathic, left heart disease, lung disease/hypoxia, chronic thromboembolism, multifactorial	698
SIADH	Small cell carcinoma of the lung	703

► EQUATION REVIEW

TOPIC	EQUATION	PAGE
Volume of distribution	$V_d = \frac{\text{amount of drug in the body}}{\text{plasma drug concentration}}$	229
Half-life	$t_{1/2} = \frac{0.7 \times V_d}{CL}$	229
Drug clearance	$CL = \frac{\text{rate of elimination of drug}}{\text{plasma drug concentration}} = V_d \times K_e$ (elimination constant)	229
Loading dose	$LD = \frac{C_p \times V_d}{F}$	229
Maintenance dose	$\text{Maintenance dose} = \frac{C_p \times CL \times \tau}{F}$	229
Therapeutic index	$TI = \text{median toxic dose/median effective dose} = TD_{50}/ED_{50}$	233

TOPIC	EQUATION	PAGE
Odds ratio (for case-control studies)	$OR = \frac{a/c}{b/d} = \frac{ad}{bc}$	258
Relative risk	$RR = \frac{a/(a+b)}{c/(c+d)}$	258
Attributable risk	$AR = \frac{a}{a+b} - \frac{c}{c+d}$	258
Relative risk reduction	$RRR = (ARC - ART)/ARC$	258
Absolute risk reduction	$ARR = \frac{c}{c+d} - \frac{a}{a+b}$	258
Number needed to treat	$NNT = 1/ARR$	258
Number needed to harm	$NNH = 1/AR$	258
Likelihood ratio +	$LR+ = \text{sensitivity}/(1 - \text{specificity}) = \text{TP rate}/\text{FP rate}$	259
Likelihood ratio –	$LR- = (1 - \text{sensitivity})/\text{specificity} = \text{FN rate}/\text{TN rate}$	259
Sensitivity	$\text{Sensitivity} = \text{TP} / (\text{TP} + \text{FN})$	260
Specificity	$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP})$	260
Positive predictive value	$PPV = \text{TP} / (\text{TP} + \text{FP})$	260
Negative predictive value	$NPV = \text{TN} / (\text{TN} + \text{FN})$	260
Cardiac output	$CO = \frac{\text{rate of } O_2 \text{ consumption}}{(\text{arterial } O_2 \text{ content} - \text{venous } O_2 \text{ content})}$ $CO = \text{stroke volume} \times \text{heart rate}$	290
Mean arterial pressure	$MAP = CO \times \text{total peripheral resistance (TPR)}$ $MAP (\text{at resting HR}) = \frac{2}{3} DBP + \frac{1}{3} SBP = DBP + \frac{1}{3} PP$	290
Stroke volume	$SV = EDV - ESV$	290
Ejection fraction	$EF = \frac{SV}{EDV} = \frac{EDV - ESV}{EDV}$	290
Resistance	$\text{Resistance} = \frac{\text{driving pressure } (\Delta P)}{\text{flow } (Q)} = \frac{8\eta (\text{viscosity}) \times \text{length}}{\pi r^4}$	291
Capillary fluid exchange	$J_v = \text{net fluid flow} = K_f[(P_c - P_i) - \sigma(\pi_c - \pi_i)]$	301
Reticulocyte production index	$RPI = \% \text{ reticulocytes} \times \left(\frac{\text{actual Hct}}{\text{normal Hct}} \right) / \text{maturation time}$	423
Renal clearance	$C_x = (U_x V)/P_x$	600
Glomerular filtration rate	$C_{\text{inulin}} = \text{GFR} = U_{\text{inulin}} \times V/P_{\text{inulin}}$ $= K_f [(P_{\text{GC}} - P_{\text{BS}}) - (\pi_{\text{GC}} - \pi_{\text{BS}})]$	600
Effective renal plasma flow	$eRPF = U_{\text{PAH}} \times \frac{V}{P_{\text{PAH}}} = C_{\text{PAH}}$	600
Filtration fraction	$FF = \frac{\text{GFR}}{\text{RPF}}$	601

TOPIC	EQUATION	PAGE
Fractional excretion of sodium	$Fe_{Na^+} = V \times U_{Na} / GFR \times P_{Na} = P_{Cr} \times U_{Na} / U_{Cr} \times P_{Na}$	602
Henderson-Hasselbalch equation (for extracellular pH)	$pH = 6.1 + \log \frac{[HCO_3^-]}{0.03 P_{CO_2}}$	609
Winters formula (for predicted respiratory compensation for simple metabolic acidosis)	$P_{CO_2} = 1.5 [HCO_3^-] + 8 \pm 2$	609
Anion gap	$Na^+ - (Cl^- + HCO_3^-)$	610
Physiologic dead space	$V_D = V_T \times \frac{P_{aCO_2} - P_{ECO_2}}{P_{aCO_2}}$	682
Pulmonary vascular resistance	$PVR = \frac{P_{pulm\ artery} - P_{L\ atrium}}{Cardiac\ output}$	684
Alveolar gas equation	$PAO_2 = PIO_2 - \frac{P_{aCO_2}}{RQ} = 150\text{ mm Hg}^a - P_{aCO_2} / 0.8$	685

► EASILY CONFUSED MEDICATIONS

DRUG	CLINICAL USE/MECHANISM OF ACTION
Amiloride	HTN, K ⁺ -sparing diuretic (blocks epithelial Na ⁺ channels in the late distal convoluted tubule and collecting duct)
Amiodarone	Ventricular arrhythmia, atrial fibrillation, supraventricular tachycardia (K ⁺ channel blocker; class III antiarrhythmic)
Amlodipine	HTN, angina (dihydropyridine Ca ²⁺ channel blocker)
Benzotropine	Parkinson disease, extrapyramidal symptoms (cholinergic antagonist)
Bromocriptine	Parkinson disease, hyperprolactinemia (dopamine agonist; rarely used)
Buspirone	Generalized anxiety disorder (partial 5-HT _{1A} -receptor agonist)
Bupropion	Depression, smoking cessation (norepinephrine-dopamine reuptake inhibitor)
Cimetidine	Gastritis, peptic ulcer (H ₂ -receptor antagonist)
Cetirizine	Allergy (2nd-generation H ₁ -receptor antagonist)
Chloramphenicol	Antibiotic (blocks 50S subunit)
Chlordiazepoxide	EtOH withdrawal (↑ frequency of Cl ⁻ channel opening in GABAergic neuron membranes; long-acting benzodiazepine)
Chlorpromazine	Schizophrenia (typical antipsychotic, D2-receptor blockade)
Chlorpropamide	Diabetes (1st-generation sulfonylurea)
Chlorpheniramine	Allergy (1st-generation H ₁ -receptor antagonist)
Chlorthalidone	HTN, edema (inhibits Na ⁺ and Cl ⁻ reabsorption in the distal convoluted tubule; thiazide diuretic)
Clozapine	Schizophrenia (atypical antipsychotic, D2-receptor partial agonist)
Clomipramine	Depression, anxiety, chronic pain (tricyclic antidepressant)

DRUG	CLINICAL USE/MECHANISM OF ACTION
Clomiphene	Infertility due to anovulation (selective estrogen receptor modulator in hypothalamus)
Clonidine	Hypertensive urgency, ADHD (α_2 -receptor agonist)
Doxepin	Depression, anxiety, bipolar disorder (tricyclic antidepressant)
Doxazosin	BPH, HTN (α_1 -receptor antagonist)
Eplerenone	HTN, K^+ -sparing diuretic (selective mineralocorticoid receptor antagonist)
Propafenone	Ventricular arrhythmia, paroxysmal atrial fibrillation/flutter or supraventricular tachycardia (Na^+ channel blocker; class Ic antiarrhythmic)
Fluoxetine	Depression (selective serotonin reuptake inhibitor)
Fluphenazine	Schizophrenia (typical antipsychotic, D_2 -receptor antagonist)
Mifepristone	Pregnancy termination (progesterone receptor antagonist)
Misoprostol	Used with mifepristone for pregnancy termination (synthetic PGE_1 analog)
Naloxone	Opioid overdose (opioid receptor antagonist)
Naltrexone	Alcohol and opioid use disorder (opioid receptor antagonist)
Nitroprusside	Hypertensive emergency ($\uparrow$ cGMP/NO)
Nitroglycerin	Angina ($\uparrow$ cGMP/NO)
Omeprazole	GERD (inhibits H^+/K^+ -ATPase in parietal cells)
Ketoconazole	Antifungal (inhibits fungal sterol synthesis)
Aripiprazole	Schizophrenia (atypical antipsychotic, D_2 partial agonist)
Anastrozole	ER $\oplus$ breast cancer in postmenopausal women (aromatase inhibitor)
Rifaximin	Hepatic encephalopathy (inhibits DNA-dependent RNA polymerase $\rightarrow$ $\downarrow$ ammoniagenic bacteria)
Rifampin	Antituberculous drug/antimicrobial (inhibits DNA-dependent RNA polymerase)
Sertraline	Depression, PTSD (selective serotonin reuptake inhibitor)
Selegiline	Parkinson disease (MAO-B inhibitor)
Trazodone	Insomnia (blocks 5-HT ₂ , α_1 -adrenergic, and H_1 receptors; also weakly inhibits 5-HT reuptake)
Tramadol	Chronic pain (weak opioid agonist)
Varenicline	Smoking cessation (nicotinic ACh receptor partial agonist)
Venlafaxine	Depression, diabetic neuropathy (serotonin-norepinephrine reuptake inhibitor)